

NAVIGATE + FILES

pwd	current directory
ls / ls -la	list files / detailed + hidden
cd dir / cd .. / cd ~	move / parent / home
mkdir dir / mkdir -p a/b	create directory / parent path
touch file	create empty file
cp a b / cp -r d1 d2	copy file / directory
mv a b	move or rename
rm file / rm -r dir	delete file / directory
cat file	print file
less file	scroll large file
head -n 10 file	first 10 lines
tail -n 10 file	last 10 lines
tail -f app.log	follow live log
echo "text" > f	overwrite file
echo "text" >> f	append to file
wc -l file	count lines
diff a b	compare files

NETWORKING

ip addr	IP + interfaces
ping host	reachability/latency
netstat -tulnp	list listening ports/connections
traceroute host	route/hops to host
nslookup domain	DNS lookup
telnet host port	test TCP port
mtr host	continuous ping + route
nmap host	scan host/ports

SYSTEM + PACKAGES + PROCESSES

whoami	current user
hostname	machine name
uname -a	kernel/system info
history	command history
clear	clear terminal
date	date/time
uptime	uptime + load
free -h	RAM usage
df -h	filesystem disk usage
du -sh dir	directory size
ps aux	running processes
top	live process/resource view
kill PID / kill -9 PID	stop / force-stop process
sudo command	run with admin privilege
apt update	refresh Ubuntu package list
apt install pkg	install package
apt remove pkg	remove package

SEARCH + TEXT PROCESSING

grep "error" file	find matching lines
grep -i "error" file	case-insensitive
grep -r "text" dir/	recursive search
find /path -name "*.log"	find by filename
find /path -type f	find files
awk '{print \$1}' file	print first column
awk -F: '{print \$1}' /etc/passwd	custom delimiter
sed 's/old/new/' file	replace first match/line
sed 's/old/new/g' file	replace all matches/line

PERMISSIONS + USERS + GROUPS

ls -l	see owner/group + rwx
chmod 755 file	u=rwx, g=rx, o=rx
chmod +x script.sh	add execute permission
chown user file	change owner
chown user:group file	change owner + group
chgrp group file	change group
useradd -m user	create user + home
passwd user	set/change password
su - user	switch/login as user
userdel -r user	delete user + home
groupadd dev	create group
gpasswd -a user dev	add user to group
groups user	show user groups
ln source hardlink	hard link
ln -s target link	symbolic link

ARCHIVE + TRANSFER + REMOTE

zip -r out.zip dir	zip a directory
unzip out.zip	extract zip
tar -czvf a.tar.gz dir	create tar.gz
tar -xzvf a.tar.gz	extract tar.gz
ssh user@host	remote shell
ssh -i key.pem user@host	SSH with private key
scp file user@host:/path	copy file to remote
scp -r dir user@host:/path	copy directory to remote
rsync -avz src/ user@host:/dst/	efficient sync/copy
wget URL	download file
curl URL	request/fetch URL

DISKS + MOUNT + LVM

lsblk	list disks/partitions
fdisk -l	disk/partition details
mount /dev/x /mnt	mount filesystem
umount /mnt	unmount
pvccreate /dev/x	make LVM physical volume
pvdiskdisplay	show physical volumes
vgcreate vg1 /dev/x	create volume group
vgdisplay	show volume groups
lvcreate -L 5G -n lv1 vg1	create logical volume
lvdisplay	show logical volumes
mkfs.ext4 /dev/vg1/lv1	format logical volume
lvextend -L +5G /dev/vg1/lv1	extend logical volume
resize2fs /dev/vg1/lv1	grow ext4 filesystem